FitPhone - Project Proposal

Introduction

This document describes the idea and plan for the FitPhone project. The project explores how artificial intelligence can be used to analyze smartphone and social media usage behavior and detect patterns that may relate to health or wellbeing.

During an earlier meeting with the stakeholder the idea was suggested that the team could create TikTok or other social media accounts to collect data. However, the team decided not to follow this direction because of personal and legal concerns. This was also discussed with a teacher, who confirmed that collecting data in this way would not be appropriate for this project. Because of this the team decided that it would be better to work with existing datasets or other forms of real data that can be used safely and responsibly.

The stakeholder agreed with this approach. It was also agreed that by next Monday the team will present a project proposal. In the meantime, the stakeholder will look into whether it is possible to gather suitable data that the team could potentially use in the project.

Based on this advice, we decided to explore how AI could analyze smartphone usage patterns and relate them to possible health indicators such as stress, sleep problems, or unhealthy usage habits. The idea behind FitPhone is that it could eventually function as a tool that helps users better understand how their phone usage might affect their wellbeing.

Purpose or Goal

Many people spend a large amount of time on their phones every day, especially on social media. Long screen time, irregular phone usage, or using a phone late at night can sometimes be linked to things like poor sleep, stress, or reduced wellbeing.

The problem is that people often do not realize how their phone habits affect them. While phones collect a lot of usage data, this information is rarely analyzed in a way that gives meaningful insights about behavior or potential health risks.

The goal of the FitPhone project is to explore whether AI can analyze smartphone usage data and detect patterns that may be related to wellbeing. By working with existing datasets that include information about screen time and phone usage behavior, we want to see whether machine learning models can find connections between usage patterns and possible health indicators.

In addition to usage data, another possible direction is analyzing text or content data from social media platforms. Some datasets contain posts or written statements that are labeled with mental health indicators. AI models can analyze language patterns in these texts to detect signals that may relate to emotional distress or mental wellbeing.

The main benefit of this project is gaining a better understanding of how AI can analyze both behavioral data and text data, and how smartphone usage habits or online communication might relate to wellbeing.

Deliverables

The project will result in a working prototype that demonstrates how AI can analyze smartphone and social media data.

The system will mainly use existing datasets with information about phone usage, such as screen time and frequency of use. The AI model will analyze this data to find patterns that may relate to wellbeing. If suitable datasets are available, we may also analyze text-based datasets with social media posts or written statements.

The project will be developed in several sprints with the following deliverables.

Sprint 1 - Ideation and Research

- Brainstorm Document
- Lotus Blossom
- Empathy Maps
- Project Plan
- Concept Document
- Personas
- Competitive Analysis
- Best, Good and Bad Practices
- Expert Interview

Sprint 2 - Development

- Git Repository
- Proof of Concept
- First Prototype

Sprint 3 - Testing and Iteration

- Usability Testing
- A/B Testing
- User Observation
- User Survey

Sprint 4 - Finalization

- Final Working Prototype
- Client Meeting Notes
- Teacher Validation (Feedback)

These deliverables show the progress of the project from the first ideas to the final working prototype.

Requirements

To develop the FitPhone project, access to suitable datasets is needed. These datasets should contain information about smartphone or social media usage behavior such as screen time, frequency of use, or related indicators.

Public datasets will be used for this purpose, including datasets that explore the relationship between screen time and mental health.

If available, the project may also use datasets containing text or social media statements that are labeled with mental health indicators. These datasets can be analyzed using natural language processing techniques to detect patterns in written communication.

Development tools such as Python and common data analysis libraries will also be required to process the data and build the machine learning models.

Feedback from teachers and project stakeholders will help guide the project and ensure the approach stays realistic for the semester challenge.

Project Team

Because the group has a mix of different strengths, including technical development, research, and design, we are considering building a simple application using Unity as a wrapper to display the insights and results from the AI model in a clear and interactive way. This approach allows us to combine the technical AI work with a visual front-end where the results can be presented in a more understandable way.

The goal of the team is to combine technical experimentation with research to better understand how behavioral data can be analyzed using AI.

Planning or Approach

The project will be developed step by step during the semester. The first sprint will focus on researching datasets and understanding what kind of smartphone usage data is available. During this sprint the team will also explore how this data can be structured and analyzed.

The second sprint will focus on preparing and cleaning the datasets so they can be used for analysis. After that, the AI model will be developed and trained to detect patterns in the usage data. If text datasets are used NLP techniques will also be explored to analyze patterns in text.

The final sprint will focus on analyzing the results, creating visualizations, and preparing the final presentation of the project.

Costs or Budget

Since FitPhone is a prototype for a school project, the costs will be neglectable. Most tools we will use, such as Python and machine learning libraries, are free and open source. The datasets we plan to use are also publicly available, so we do not expect any costs for data. Because of this, the main investment in the project will be time spent on research, development, and testing rather than money.

Conclusion

This proposal describes the plan for the FitPhone project. The goal of the project is to explore how AI can analyze smartphone and social media usage and find patterns related to wellbeing. The project will result in a prototype that provides useful insights based on this data.

During discussions with the client, it became clear that the AI direction and the exact datasets for FitPhone are still being explored. Because of this, we also researched publicly available datasets that could be used for the project. If these datasets are used, we will follow the approach described in this proposal.

Possible Datasets

Usage / Behavioral Data

Impact of Screen Time on Mental Health
Social Media and Mental Health

Text / Content Data

Depression : Reddit Dataset
Sentiment Analysis for Mental Health